

Blood

- 1. Definitions**
- 2. Blood components**
- 3. Hematopoiesis**
- 4. Hemostasis**
- 5. Blood typing (*in lab*)**

Definitions

1. **Hematology** – *study of blood*
2. **Hematopoiesis** – *formation of blood cells
(red bone marrow)*
3. **Hemoglobin** – *transport O_2 (CO_2 , buffer)*
4. **Hematocrit** - *% of total blood volume that is
RBCs*
5. **Hemostasis** – *“blood stop”; blood clotting*
6. **Hemostat** – *instrument used in operating
room to stop blood flow*

Blood Components

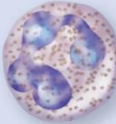
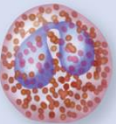
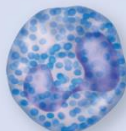
plasma + blood cells

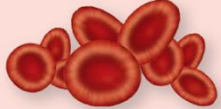
Hematocrit: Percent RBCs

- Average: 44%
- Female: 36%-46%
- Male: 41%-53%



Plasma (55% of whole blood)		
Water 92% by weight	Proteins 7% by weight Albumins 58% Globulins 37% Fibrinogen 4% Regulatory proteins <1%	Other solutes 1% by weight Electrolytes Nutrients Respiratory gases Waste products

Buffy Coat (<1% of whole blood)	
Platelets 150–400 thousand per cubic mm 	Leukocytes 4.5–11 thousand per cubic mm  Neutrophils 50–70%  Lymphocytes 20–40%  Monocytes 2–8%  Eosinophils 1–4%  Basophils 0.5–1%

Erythrocytes (44% of whole blood)
Erythrocytes 4.2–6.2 million per cubic mm 

Formed elements

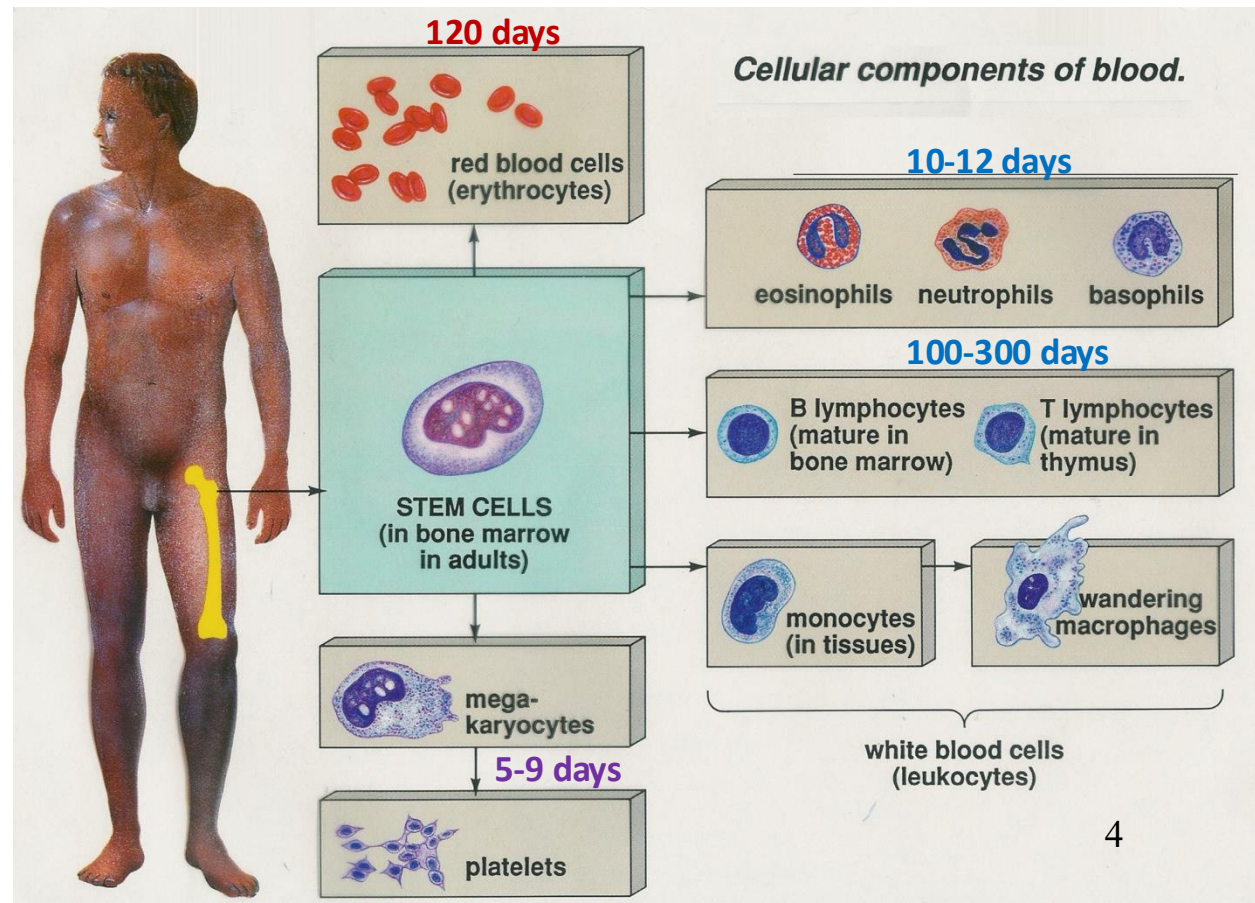
Blood functions:

- **Plasma** – transport nutrients, waste, clotting factors, immune proteins, etc.
- **Thrombocytes/platelets** - clotting
- **White blood cells/WBCs** – immune defense
- **Erythrocytes/RBCs** – gas transport

Hematopoiesis: formation of blood cells; ongoing

- **Erythropoiesis:** formation of red blood cells; kidneys release erythropoietin as blood O_2 decreases.
- **Leukopoiesis:** formation of white blood cells
- **Thrombopoiesis:** formation of platelets

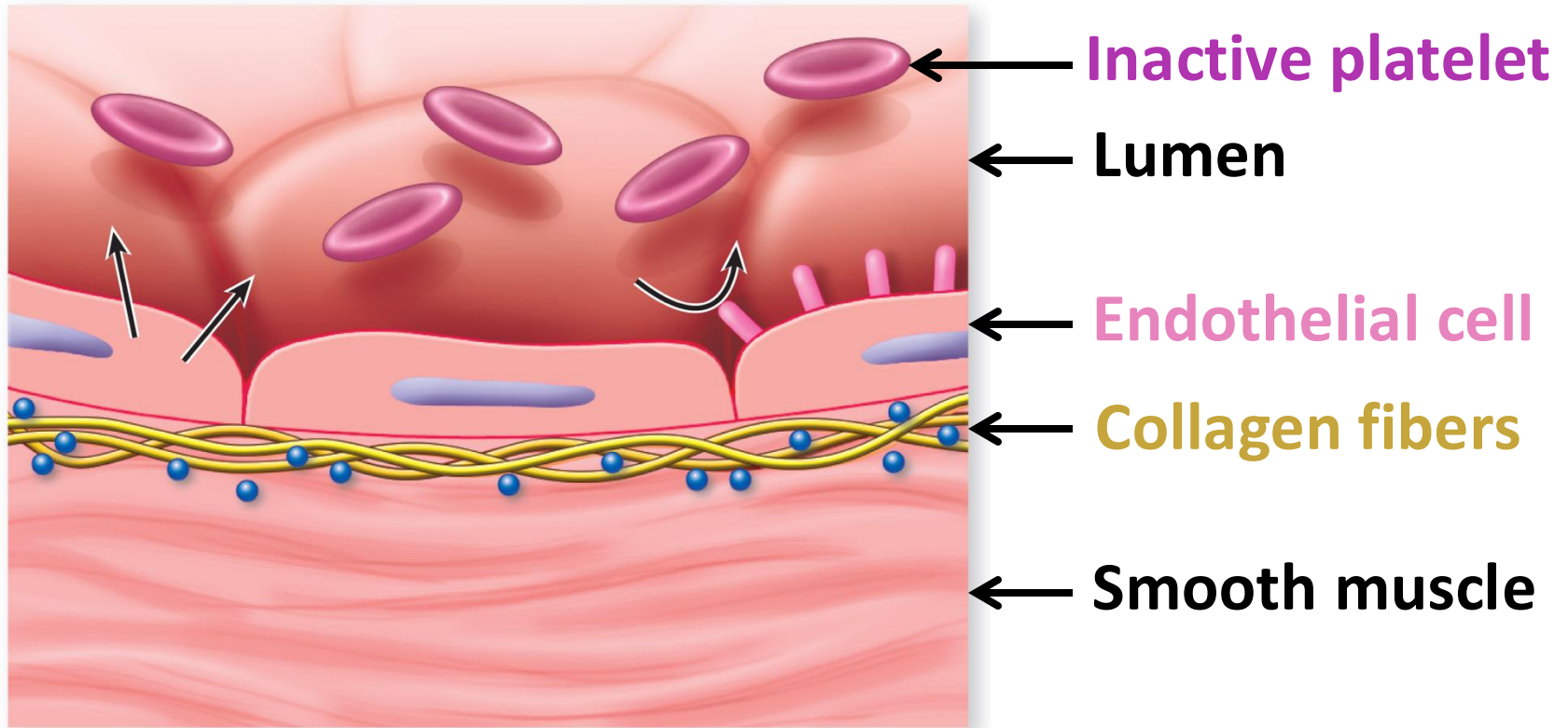
poiesis = to make



Blood

1. Definitions
2. Blood components
3. Hematopoiesis
4. Hemostasis
5. Blood typing (*in lab*)

Normal Blood Vessel Wall



Vascular injury → exposes collagen → initiates **hemostasis**

Hemostasis: 3 phases

“stop blood”

1. Vasoconstriction

→ decreases blood loss

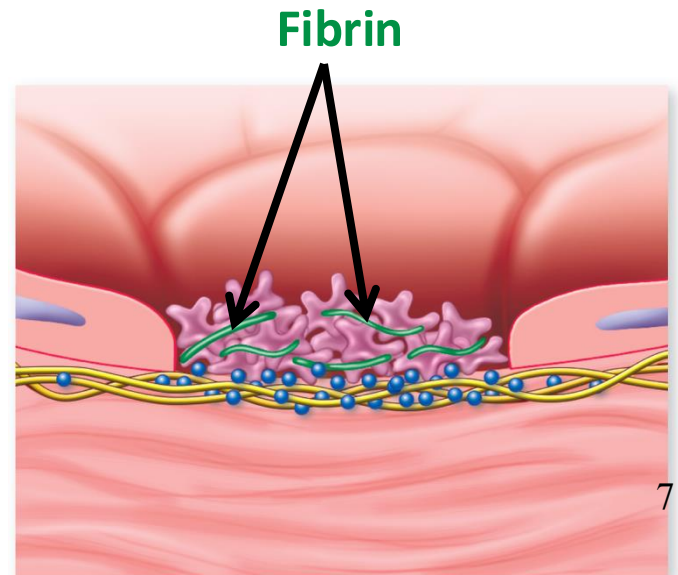
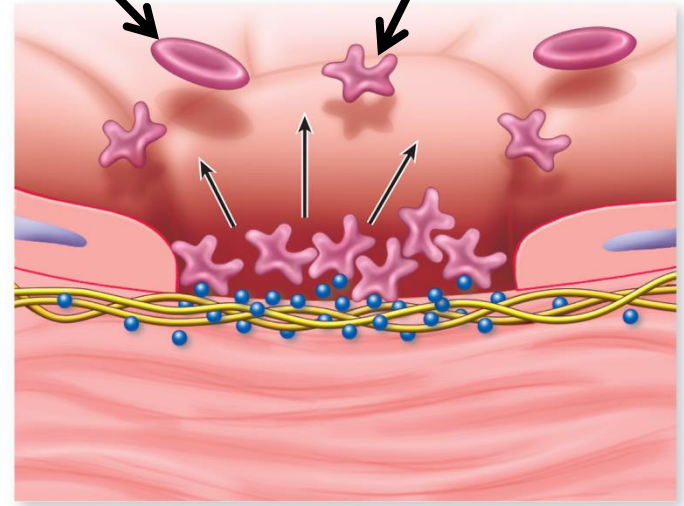
2. Platelet plug formation

- Exposed collagen → activates platelets that...
 - Release factors
 - Becomes sticky
 - Forms loose plug

3. Coagulation: clot forms

- *clotting factors* convert soluble fibrinogen into insoluble fibrin
 - Binds platelets
 - Forms stable clot

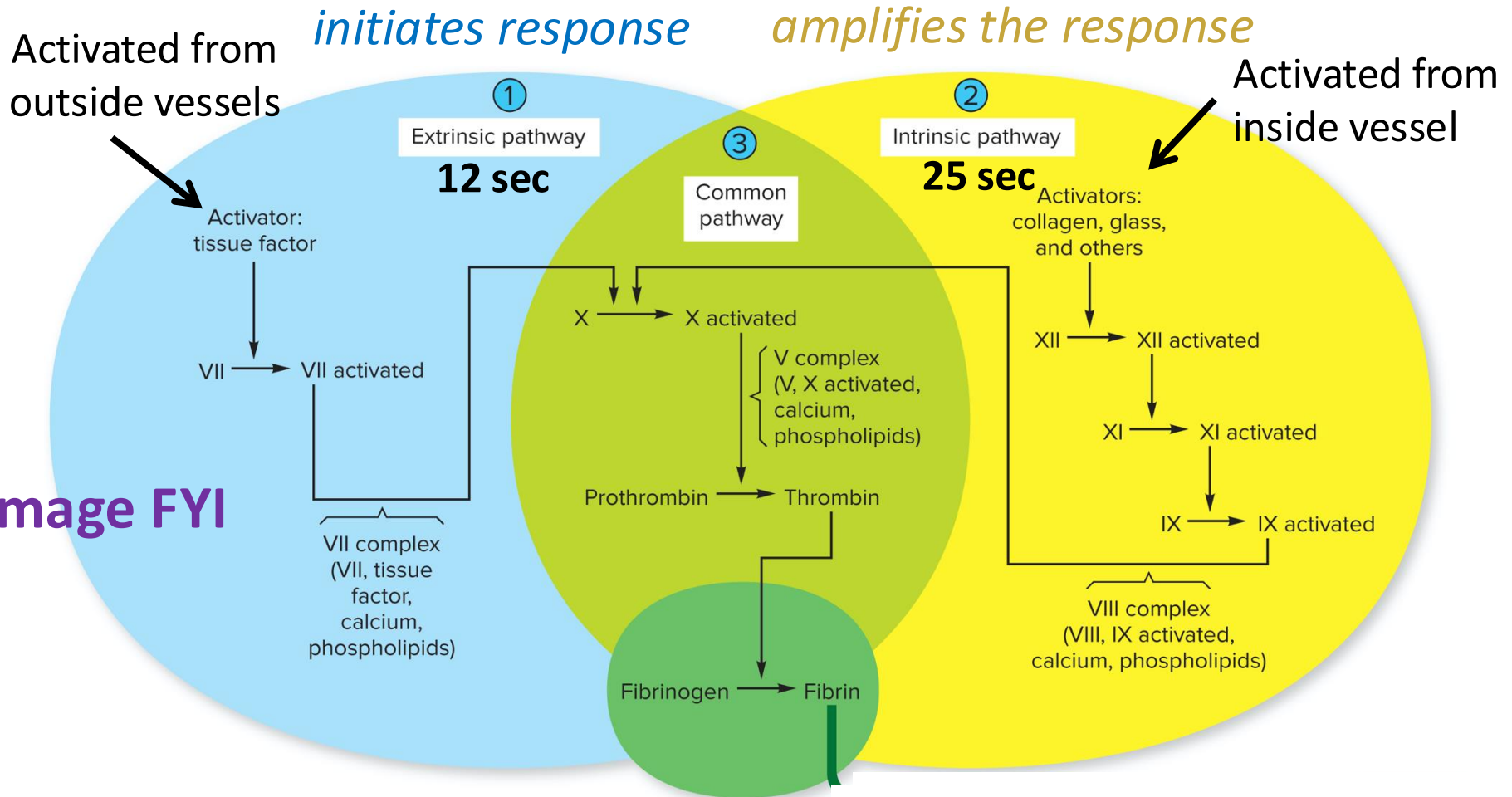
Unactivated platelet Activated platelet



Coagulation Cascade: 3 Pathways

1. Extrinsic pathway:
 - Activated by: tissue factors released with tissue injury
 - **Result: activate Factor X → activates Common Pathway.**
 - Faster: 12 seconds
2. Intrinsic pathway:
 - Activated by: factors within the blood.
 - **Result: activate Factor X → activates Common Pathway.**
 - Slower: 25 seconds
3. Common pathway: activated by **Factor X.**

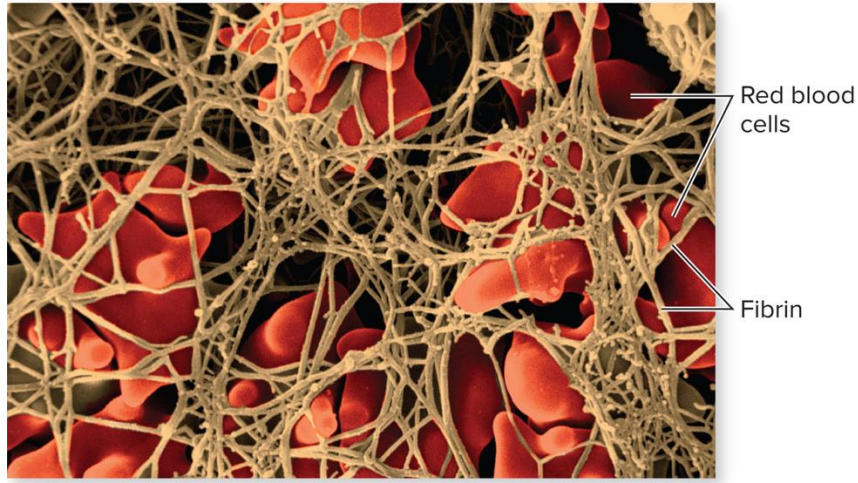
Clotting factor cascade produces **fibrin** (Ca^{2+} required)



Clotting factors (inactive blood proteins) are made by liver; once activated → → → activated factor X → thrombin → **fibrin**

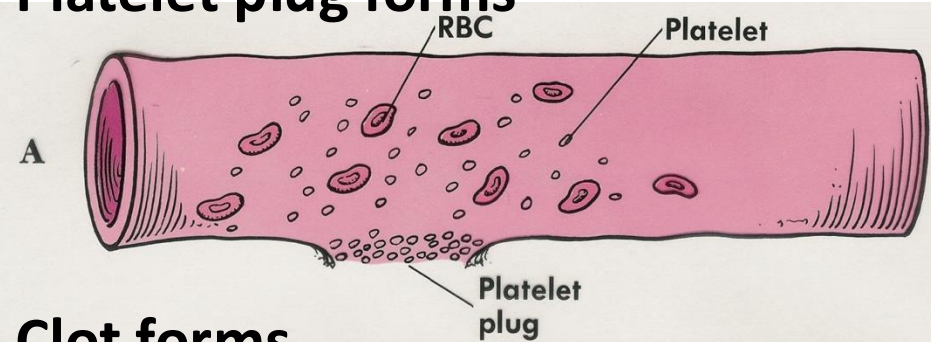
Hemophilia: genetic bleeding disorder, missing any clotting factor

Removing Clot After Injury Heals

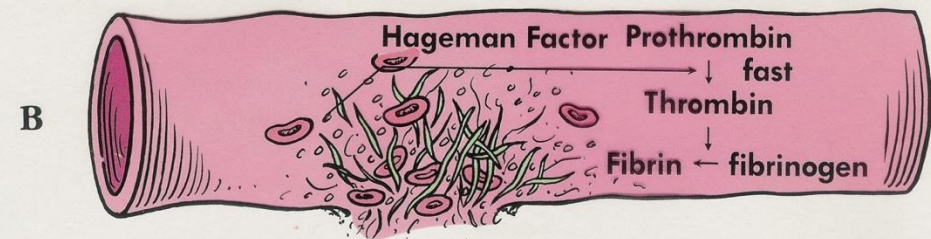


Clot: fibrin + platelet + trapped RBCs

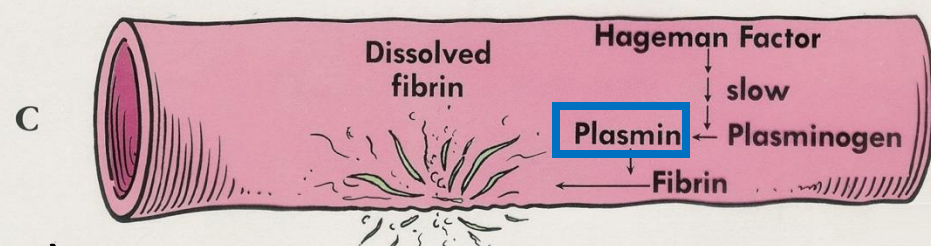
Platelet plug forms



Clot forms



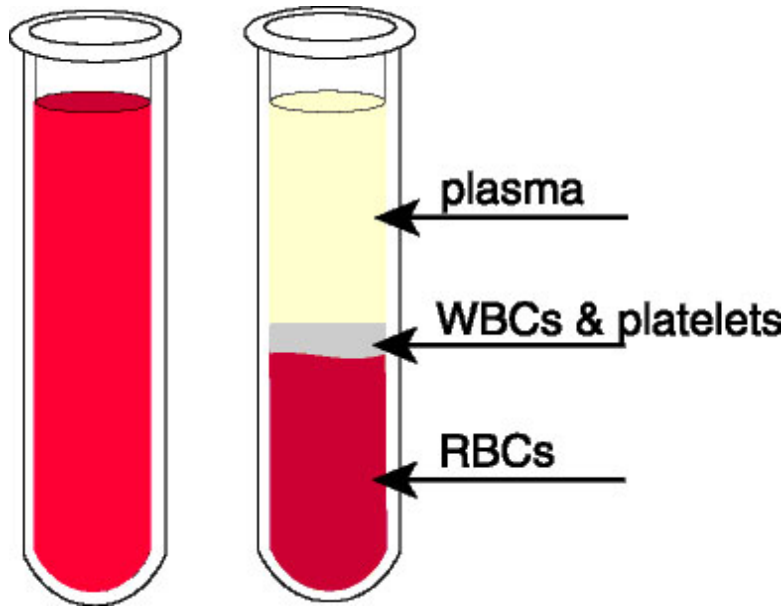
Plasmin dissolves clot



FYI: Blood clotting video (5:04)

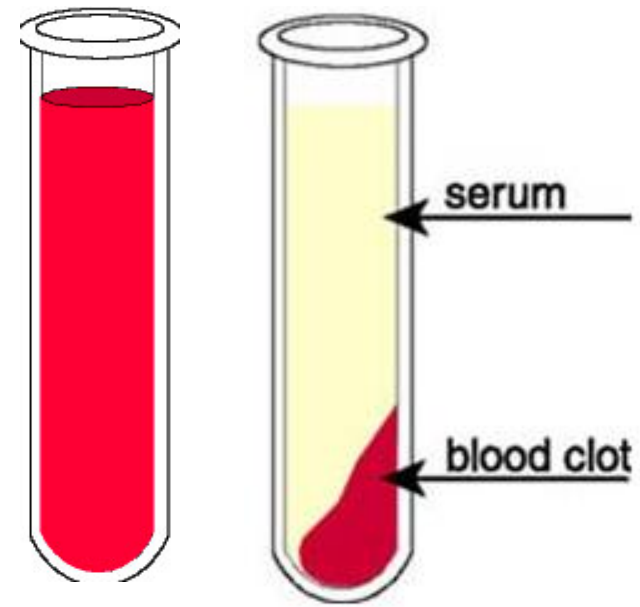
<https://www.youtube.com/watch?v=x8TLTTyyPfl>

Plasma versus Serum



Centrifuge blood

Clotting factors in plasma



**Do not centrifuge blood;
let blood stand**

Clotting factors are in blood clot

- **Serum** = plasma – clotting factors
- Glass test tubes: clot blood
- Heparin-coated glass or plastic test tubes: prevents clotting in test tubes
- Coumadin: prevents clotting in patients